



## Week 6 - Graded Assignment 6



**The due date for submitting this assignment has passed.**

Due on 2026-03-25, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

**You have last submitted on: 2026-03-25, 07:24 IST**

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

### Multiple Select Questions (MSQ):

1 point

A function  $T: V \rightarrow W$  between two vector spaces  $V$  and  $W$  is said to be a linear transformation if the following conditions hold:

- Condition 1:  $T(v_1 + v_2) = T(v_1) + T(v_2)$  for all  $v_1, v_2 \in V$ .
- Condition 2:  $T(c v) = c T(v)$  for all  $v \in V$  and  $c \in R$ .

Consider the following function:

$$T: R^2 \rightarrow R \quad T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y \neq 0 \end{cases}$$

Which of the following statements is true?

- ☐ Condition 1 holds.
- ☒ Condition 1 does not hold.
- ☒ Condition 2 holds.
- ☐ Condition 2 does not hold.

**Yes, the answer is correct.**

**Score: 1**

### Accepted Answers:

Condition 1 does not hold.

Condition 2 holds.

1 point

Suppose the matrix representation of a linear transformation  $T: R^3 \rightarrow R^3$  with respect to ordered bases  $\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$  for the domain and  $\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$  for the range, is  $I_{3 \times 3}$ , i.e., the identity matrix of order 3. Let  $A$  denote the matrix representation of the linear transformation  $T$  with respect to the standard ordered basis of  $R^3$  for both domain and range. Which of the following are true?

- ☐  $A = I_{3 \times 3}$  i.e., identity matrix of order 3.
- ☐  $A$  is a singular matrix



$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix} A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$



$$\det(A) = 1 \det(A) = 1.$$



$$\det(A) = -1 \det(A) = -1.$$



$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$$\det(A) = 1 \det(A) = 1.$$

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

*1 point*

Match the linear transformations and sets of vectors in column A with the images of those sets under the linear transformation in column B and the geometric representations of both sets in column C.

Choose the correct option from the following.



i → d → 2, ii → b → 4.



i → b → 4, ii → d → 2.



iii → a → 1, iv → c → 3



iii → a → 3, iv → c → 1

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

i → b → 4, ii → d → 2.

iii → a → 3, iv → c → 1

*1 point*

Consider a linear transformation  $S: M_2(R) \rightarrow M_2(R)$  such that  $S(A) = A^T$ . Let  $BB$  be the matrix representation of  $SS$  with respect to the ordered bases:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \left\{ \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \right\}$$

of  $M_2(R)$ . Choose the set of correct options:



The order of the matrix  $BB$  is  $2 \times 2 \times 2$ .



The order of the matrix  $BB$  is  $4 \times 44 \times 4$ .



The dimension of the row space of the the matrix  $BB$  is 4.



The dimension of the column space of the matrix  $BB$  is 3.



The nullity of the the matrix  $BB$  is 1.



The rank of the the matrix  $BB$  is 4.



$SS$  is surjective.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

The order of the matrix  $BB$  is  $4 \times 44 \times 4$ .

The dimension of the row space of the the matrix  $BB$  is 4.

The rank of the the matrix  $BB$  is 4.

$SS$  is surjective.

1 point

Let  $T: R^n \rightarrow R^m$   $T: R^n \rightarrow R^m$  be a linear transformation and let  $\{v_1, v_2, \dots, v_n\}$   $\{v_1, v_2, \dots, v_n\}$  be a basis of  $R^n$   $R^n$ . Suppose  $\{T(v_1), T(v_2), \dots, T(v_n)\}$   $\{T(v_1), T(v_2), \dots, T(v_n)\}$  is a spanning set for  $R^m$   $R^m$ . Choose all the correct statements.



$T$   $T$  is a surjective linear transformation.



$T$   $T$  is a linear isomorphism.



$m \leq nm \leq n$ .



For any basis  $\{w_1, w_2, \dots, w_n\}$   $\{w_1, w_2, \dots, w_n\}$  of  $R^n$   $R^n$ , the set  $\{T(w_1), T(w_2), \dots, T(w_n)\}$   $\{T(w_1), T(w_2), \dots, T(w_n)\}$  is a spanning set for  $R^m$   $R^m$ .



The matrix of the linear transformation  $T$   $T$  has rank equal to  $nm$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$T$   $T$  is a surjective linear transformation.

$m \leq nm \leq n$ .

For any basis  $\{w_1, w_2, \dots, w_n\}$   $\{w_1, w_2, \dots, w_n\}$  of  $R^n$   $R^n$ , the set  $\{T(w_1), T(w_2), \dots, T(w_n)\}$   $\{T(w_1), T(w_2), \dots, T(w_n)\}$  is a spanning set for  $R^m$   $R^m$ .

**Comprehension Type Question:**

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose  $x$  eggs,  $y$  (in hundreds) grams of wheat, and  $z$  (in hundreds)

grams of sugar are used as the raw materials to produce  $ax + by + cz$  number of breads. We can express this as follows:

$$T: R^3 \rightarrow R$$

$$T(x, y, z) = ax + by + cz$$

where the co-ordinates in  $R^3$  denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that  $T$  is a linear transformation.

**Use the above information for Question 6, 7 and 8.**

1 point

Choose the correct set of options from the the following. (MSQ)

☐

$$\text{Nullity}(T) = 1$$

☒

$$\text{Rank}(T) = 1$$

☒

$$\text{Nullity}(T) = 2$$

☐

$$\text{Rank}(T) = 2$$

☐

$$\text{Nullity}(T) = 3$$

☐

$$\text{Rank}(T) = 3$$

☐

$T$  is neither one to one nor onto.

☐

$T$  is one to one but not onto.

☒

$T$  is onto but not one to one.

☐

$T$  is an isomorphism.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$$\text{Rank}(T) = 1$$

$$\text{Nullity}(T) = 2$$

$T$  is onto but not one to one.

1 point

Choose the set of correct statements. (MSQ)

☐

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.

☒

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.

☒

If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.

☐

If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.

☒

If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.

If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.

If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) 20

1 point

Let  $T$  be a linear transformation from  $R^{10}$  to  $R^{12}$ . Suppose a basis for the null space of  $T$  has 33 vectors. How many linearly independent vectors are needed to form a basis for the range of  $T$ ?

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) 7

1 point

Let  $M$  be the set of all skew-symmetric matrices of order 7. Then  $M$  forms a vector space under matrix addition and scalar multiplication. Let  $T$  be a linear transformation from  $M$  to  $R$  defined by  $T(A) = 6 \operatorname{tr}(A)$ , where  $\operatorname{tr}(A) = \text{trace of } A$ . What is the nullity of  $T$ ?

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) 21.0

1 point

1 point

Let  $T: R^3 \rightarrow R^2$  be defined by  $T(x, y, z) = (10x + 9y, 2z)$ . Choose the correct options about  $T$ .

☐

The matrix of  $T$  is  $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ .

☒

The matrix of  $T$  is  $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ .

☒

A basis for the null space of  $T$  is  $\{(1, -\frac{10}{9}, 0)\}$ .

☐

A basis for the null space of  $T$  is  $\{(1, -\frac{10}{9}, 0), (0, 0, 2)\}$ .

☐

A basis for the range of  $T$  is  $\{(10, 10)\}$ .

☒

A basis for the range of  $T$  is  $\{(10, 9), (0, 2)\}$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

The matrix of  $T$  is  $\begin{bmatrix} 10 & 9 \\ 0 & 2 \end{bmatrix}$ .

A basis for the null space of  $T$  is  $\{(1, -\frac{10}{9}, 0)\}$ .